ALGORITHMS AND FLOWCHARTS

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- A typical programming task can be divided into two phases:
- ***Problem solving phase***
  - produce an ordered sequence of steps that describe solution of problem
  - this sequence of steps is called an ***algorithm***
- ***Implementation phase***
  - implement the program in some programming language

STEPS IN PROBLEM SOLVING

- First produce a general algorithm (one can use ***pseudocode***)
- Refine the algorithm successively to get step by step detailed ***algorithm*** that is very close to a computer language.
- ***Pseudocode*** is an artificial and informal language that helps programmers develop algorithms. Pseudocode is very similar to everyday English.

PSEUDOCODE & ALGORITHM

- **Example 1:** Write an algorithm to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

PSEUDOCODE & ALGORITHM

Pseudocode:

- *Input a set of 4 marks*
- *Calculate their average by summing and dividing by 4*
- *if average is below 50*

```
        Print "FAIL"
    else
        Print "PASS"
```

PSEUDOCODE & ALGORITHM

- Detailed Algorithm
- Step 1: Input M1,M2,M3,M4

```
Step 2:  GRADE ← (M1+M2+M3+M4)/4
Step 3:  if (GRADE < 50) then
             Print "FAIL"
         else
             Print "PASS"
         endif
```

THE FLOWCHART

- A graphical representation of the sequence of operations in an information system or program.
  - Information system flowcharts show how data flows from source documents through the computer to final distribution to users.
  - Program flowcharts show the sequence of instructions in a single program or subroutine. Different symbols are used to draw each type of flowchart.

THE FLOWCHART

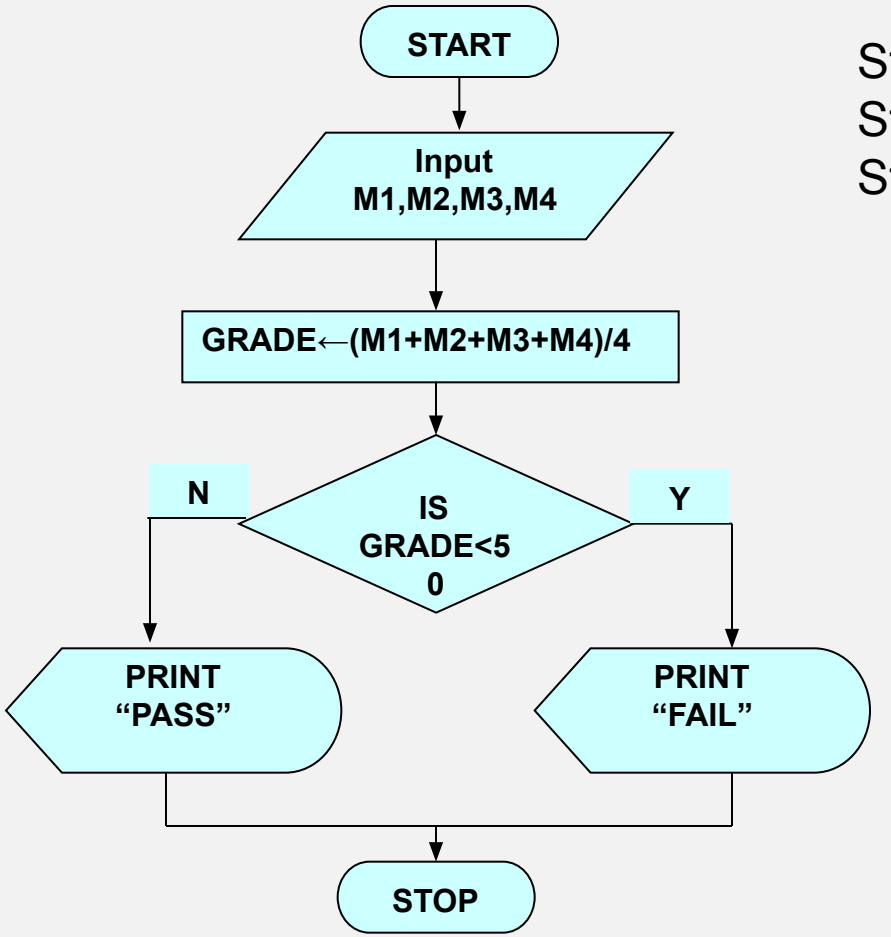
A Flowchart

- shows logic of an algorithm
- emphasizes individual steps and their interconnections
- e.g. control flow from one action to the next

FLOWCHART SYMBOLS BASIC

Name	Symbol	Use in Flowchart
Oval		Denotes the beginning or end of the program
Parallelogram		Denotes an input operation
Rectangle		Denotes a process to be carried out e.g. addition, subtraction, division etc.
Diamond		Denotes a decision (or branch) to be made. The program should continue along one of two routes. (e.g. IF/THEN/ELSE)
Hybrid		Denotes an output operation
Flow line		Denotes the direction of logic flow in the program

EXAMPLE



Step 1: Input M1,M2,M3,M4
Step 2: GRADE ← (M1+M2+M3+M4)/4
Step 3: if (GRADE <50) then
Print “FAIL”
else
Print “PASS”
endif

EXAMPLE 2

- Write an algorithm and draw a flowchart to convert the length in feet to centimeter.

Pseudocode:

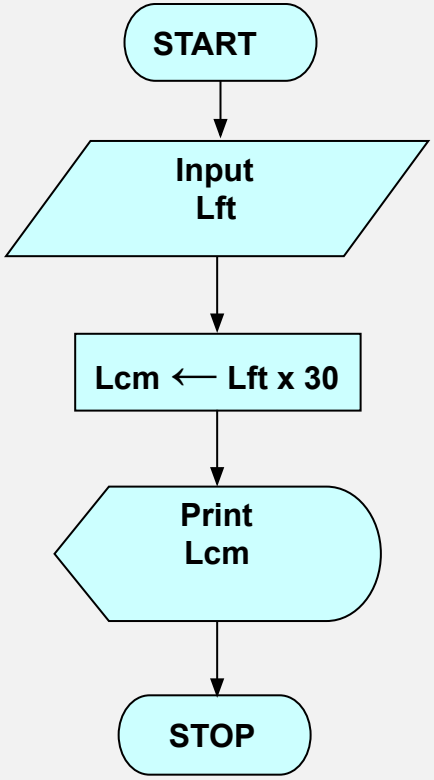
- *Input the length in feet (Lft)*
- *Calculate the length in cm (Lcm) by multiplying LFT with 30*
- *Print length in cm (LCM)*

EXAMPLE 2

Algorithm

- Step 1: Input Lft
- Step 2: Lcm ← Lft x 30
- Step 3: Print Lcm

Flowchart



EXAMPLE 3

Write an algorithm and draw a flowchart that will read the two sides of a rectangle and calculate its area.

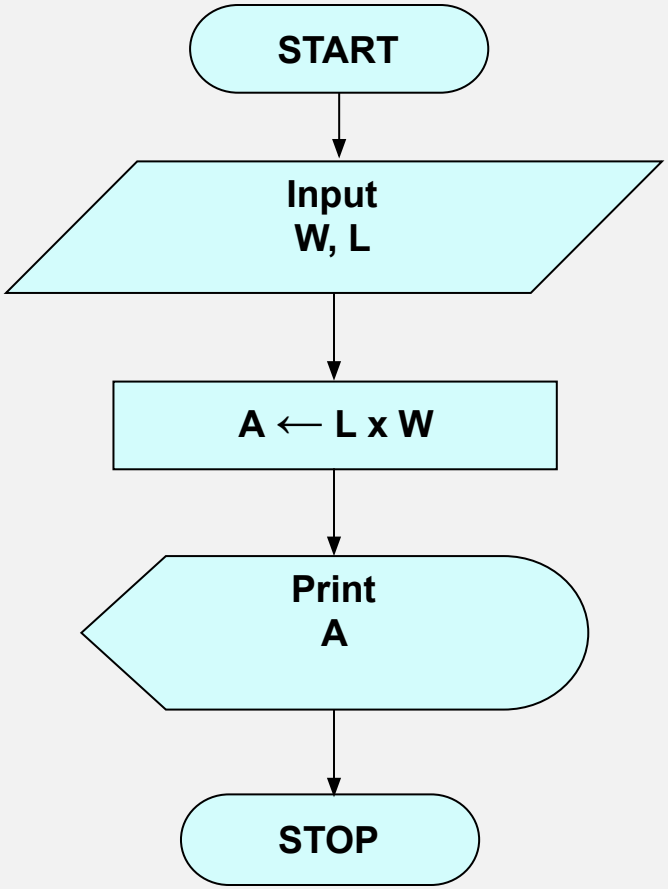
Pseudocode

- *Input the width (W) and Length (L) of a rectangle*
- *Calculate the area (A) by multiplying L with W*
- *Print A*

EXAMPLE 3

Algorithm

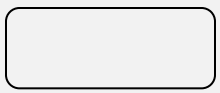
- Step 1: Input W,L
- Step 2: A ← L x W
- Step 3: Print A



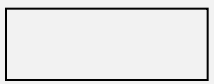
FLOWCHARTS

- Flowcharts is a graph used to depict or show a step by step solution using **symbols** which represent a task.
- The symbols used consist of geometrical shapes that are connected by **flow lines**.
- It is an alternative to pseudocoding; whereas a pseudocode description is verbal, a flowchart is graphical in nature.

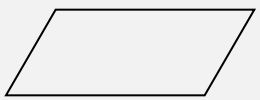
FLOWCHART SYMBOLS



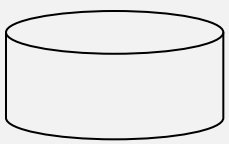
Terminal symbol - indicates the beginning and end points of an algorithm.



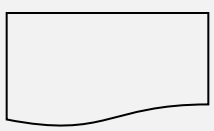
Process symbol - shows an instruction other than input, output or selection.



Input-output symbol - shows an input or an output operation.

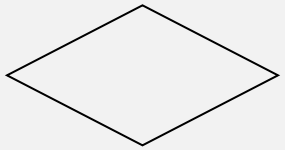


Disk storage I/O symbol - indicates input from or output to disk storage.

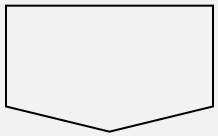


Printer output symbol - shows hardcopy printer output.

FLOWCHART SYMBOLS CONT…

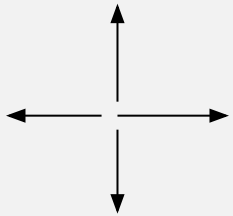


Selection symbol - shows a selection process for two-way selection.



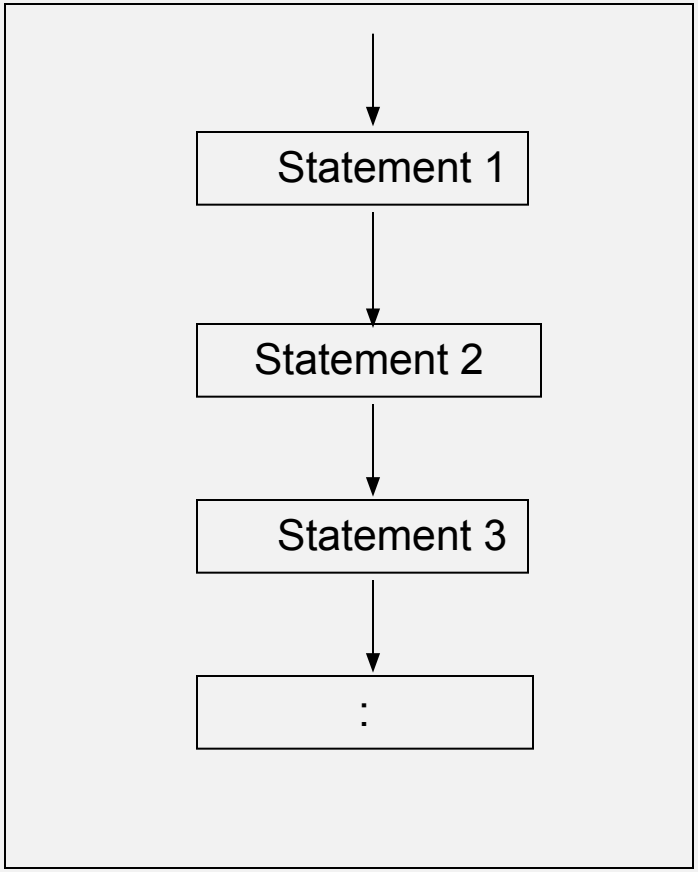
Off-page connector - provides continuation of a logical path on another page.

On-page connector - provides continuation of logical path at another point in the same page.

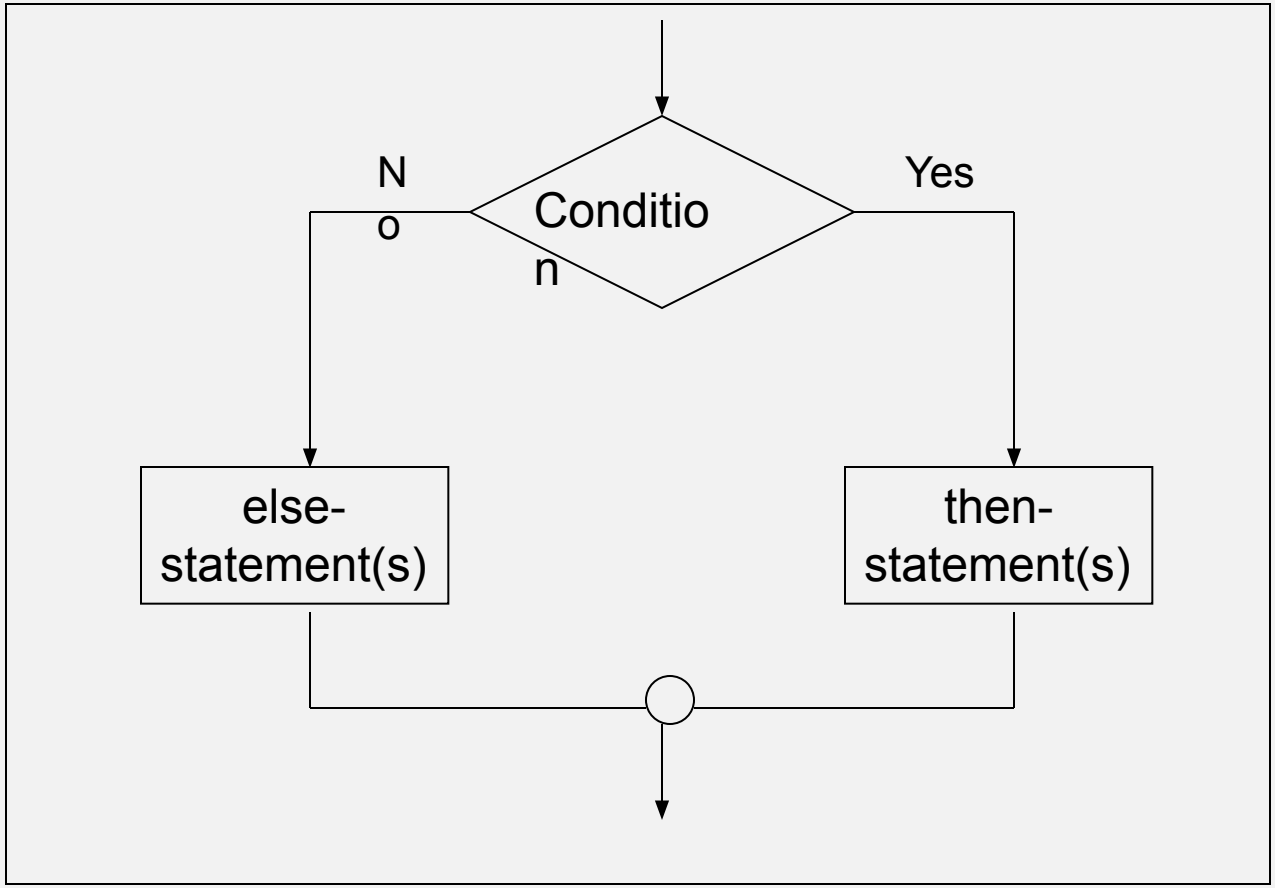


Flow lines - indicate the logical sequence of execution steps in the algorithm.

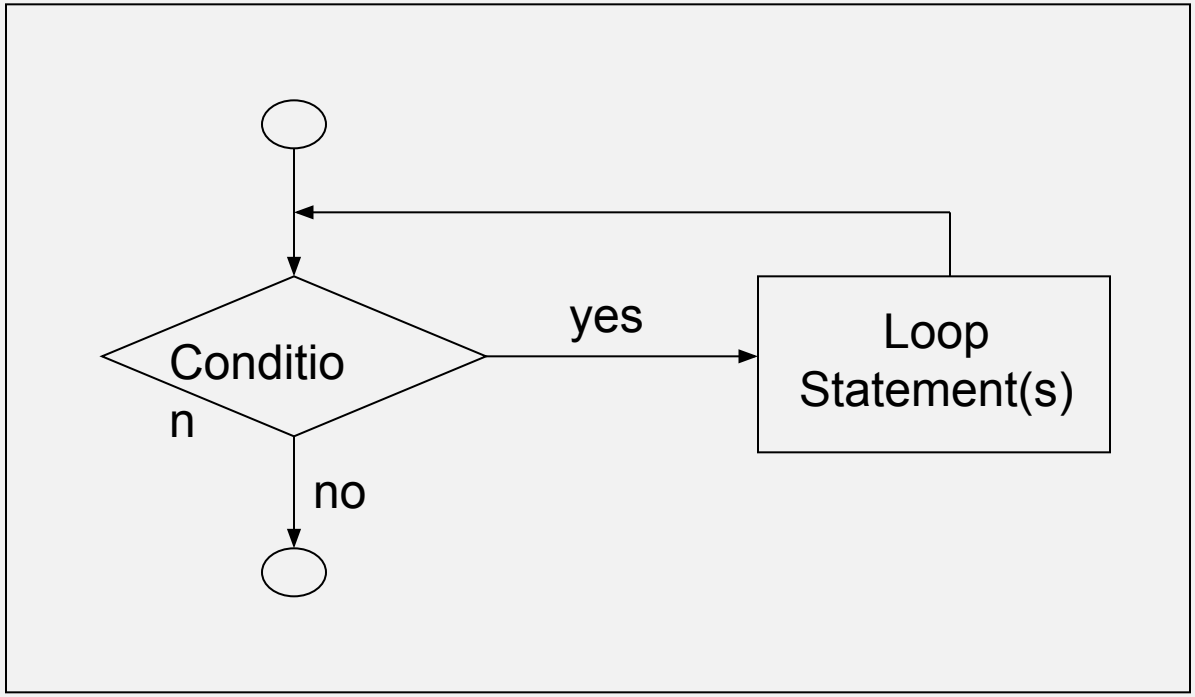
FLOWCHART – SEQUENCE CONTROL STRUCTURE



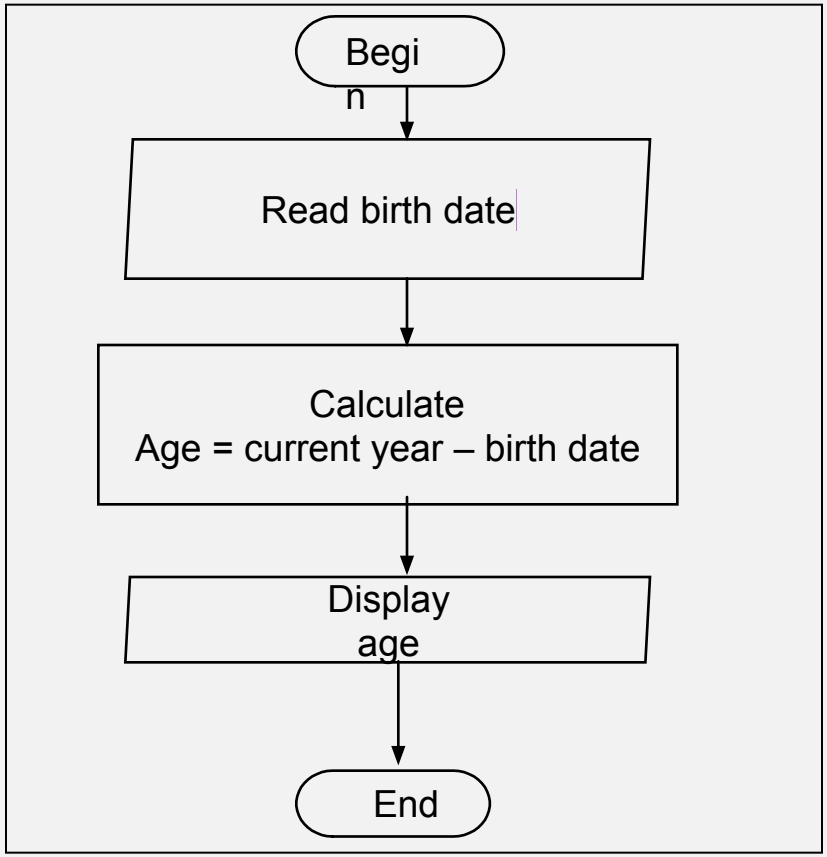
FLOWCHART – SELECTION CONTROL STRUCTURE



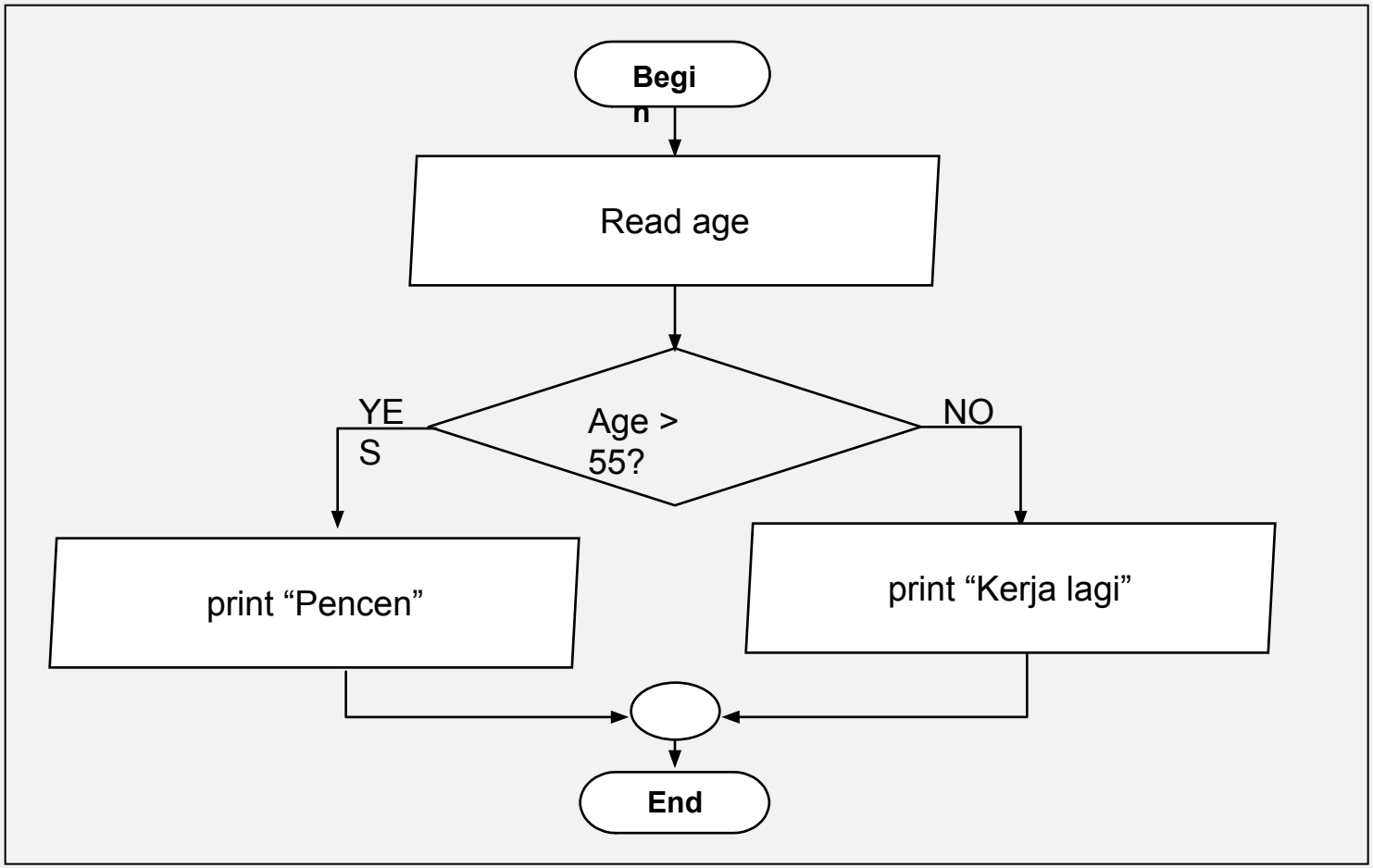
FLOWCHART – REPETITION CONTROL STRUCTURE



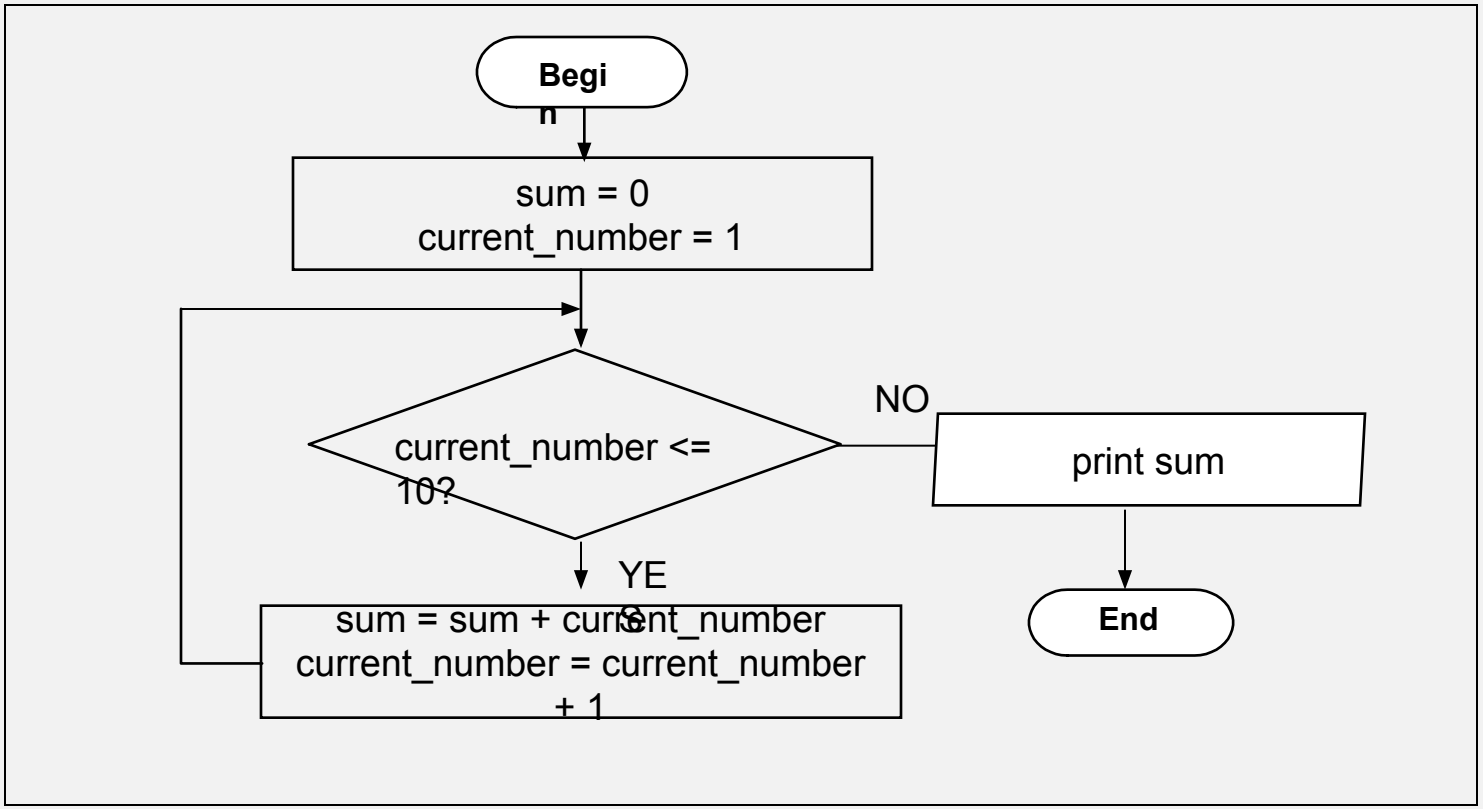
FLOWCHART – EXAMPLE 1



FLOWCHART – EXAMPLE 2



FLOWCHART – EXAMPLE 5



EXAMPLE 4

- Write an algorithm and draw a flowchart that will calculate the roots of a quadratic equation

  $ax^2 + bx + c = 0$

- Hint: **d** = sqrt ($b^2 - 4ac$), and the roots are: ***x*l** = $(-b + d)/2a$ and ***x*2** = $(-b - d)/2a$

STEP FORM ALGORITHM:

Step 1: Input a, b, c

Step 2: d ← sqrt (b x b - 4 x a x c)

Step 3: x1 ← (–b + d) / (2 x a)

Step 4: x2 ← (–b – d) / (2 x a)

Step 5: Print x1, x2

Step 6: Stop

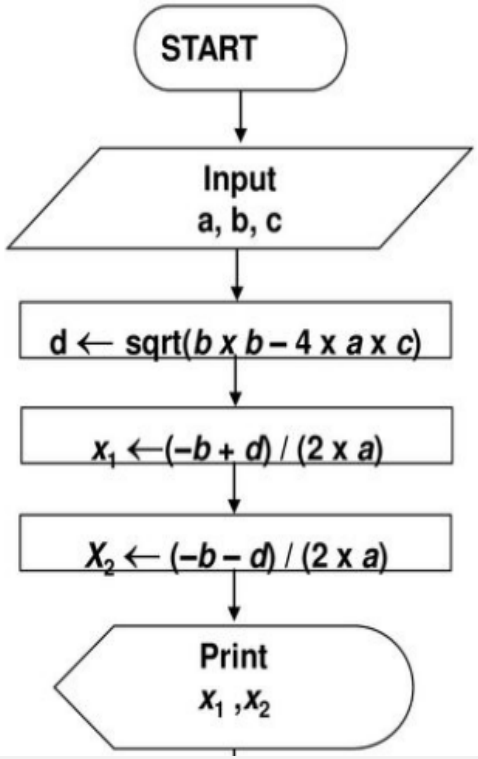
PSEUDO CODE ALGORITHM:

- Input the coefficients (a, b, c) of the quadratic equation
- Calculate d
- Calculate x1
- Calculate x2
- Print x1 and x2

FLOWCHART:

Algorithm: Quadratic Roots

- Step 1: Input a, b, c
- Step 2: $d \leftarrow \text{sqrt}\,(\,b \times b - 4 \times a \times c\,)$
- Step 3: $x1 \leftarrow (-b + d) / (2 \times a)$
- Step 4: $x2 \leftarrow (-b - d) / (2 \times a)$
- Step 5: Print $x1$, $x2$
- Step 6: Stop



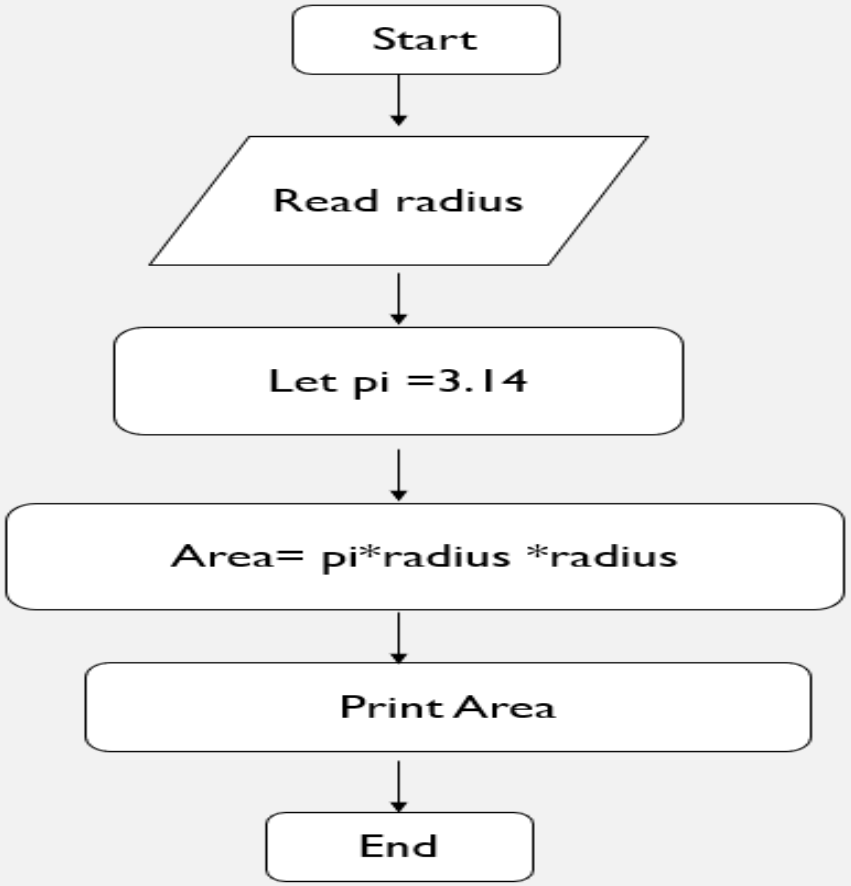
EXERCISES: ALGORITHM & FLOWCHART

2.) Create an algorithm and a flowchart that will compute the area of a circle.

ALGORITHM

Step 1: Start
Step 2: Input radius
Step 3: let pi = 3.14
Step 4: area = pi * radius * radius
Step 6: print area
Step 7: stop

FLOW CHART


Start
Read radius
Let pi =3.14
Area= pi*radius *radius
Print Area
End

For Practice

1. Create an algorithm and a flowchart that will output for g.c.d.
2. Create an algorithm and a flowchart that will output the factorial of a given number.
3. Create an algorithm and a flowchart that will accept/read two numbers and then display the bigger number.
4. Create an algorithm and a flowchart that will output the largest number among the three numbers.
5. Create an algorithm and a flowchart that will output the Fibonacci series up to a given number.
6. Create an algorithm and a flowchart that will output all the prime numbers between 2 numbers.